



ME 300: Engineering Thermodynamics

Example: Rankine Cycle

Rankine Cycle

- Rankine cycle with a mass flow of 25 kg/s and the following isentropic efficiencies: $\eta_{isen,t} = 0.93$ *turbine*
 $\eta_{isen,p} = 0.85$ *pump*
- Fill in the state table, and calculate the W_{pump} , Q_{boiler} , $W_{turbine}$, $Q_{condenser}$ and the overall thermal efficiency

State	T(K)	P(MPa)
1	290	0.15
2		20
3	990	19.5
4		0.2

Assume:
: S.C.S.
: steady-flow
: quasi-ef

$$\Delta ke = \Delta pe = 0$$

State 1 and 2

State 1: $P_1 = 0.15 \text{ MPa}$
 $T_1 = 290 \text{ K}$ } D.1

$P_{\text{sat}} = 1.92 \text{ kPa}$, $P_1 > P_{\text{sat}} \rightarrow \text{liquid}$

$\Rightarrow \text{NIST}$ $\boxed{h_1 = 70.7 \text{ kJ/kg}}$
 $s_1 = 0.25 \text{ kJ/kg}\cdot\text{K}$

State 2s: $P_2 = 20 \text{ MPa}$
 $s_{2,s} = s_1 = 0.25 \text{ kJ/kg}\cdot\text{K}$ } Table D.2 $\rightarrow s_{2,s} < s_f \rightarrow \text{liquid}$

$\Rightarrow \text{NIST}$ $T_{2s} = 290.26 \text{ K}$
 $h_{2s} = 90.654 \text{ kJ/kg}$

State 2: $\eta_P = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} = \frac{h_{2,s} - h_1}{h_2 - h_1}$

$$h_2 = \frac{h_{2,s} - h_1}{\eta_P} + h_1 = \frac{90.654 - 70.7}{0.85} + 70.7 \Rightarrow \boxed{h_2 = 94.175 \text{ kJ/kg}}$$

State 3 and 4

State 3 : $P_3 = 19.5 \text{ MPa}$
 $T_3 = 990 \text{ K}$ } D,2 $T_3 > T_{\text{sat}} \rightarrow \text{vapor}$

NIST $\Rightarrow$ $\boxed{h_3 = 3854.9 \frac{\text{kJ}}{\text{kg}}}$
 $\boxed{s_3 = 6.858 \frac{\text{kJ}}{\text{kg-K}}}$

State 4s : $P_4 = 0.2 \text{ MPa}$
 $s_{4,s} = s_3 = 6.858 \frac{\text{kJ}}{\text{kg-K}}$ } D,2 $\rightarrow \text{mixture}$

$x_{4,s} = \frac{s_{4,s} - s_f}{s_g - s_f} \rightarrow x_{4,s} = 0.952$

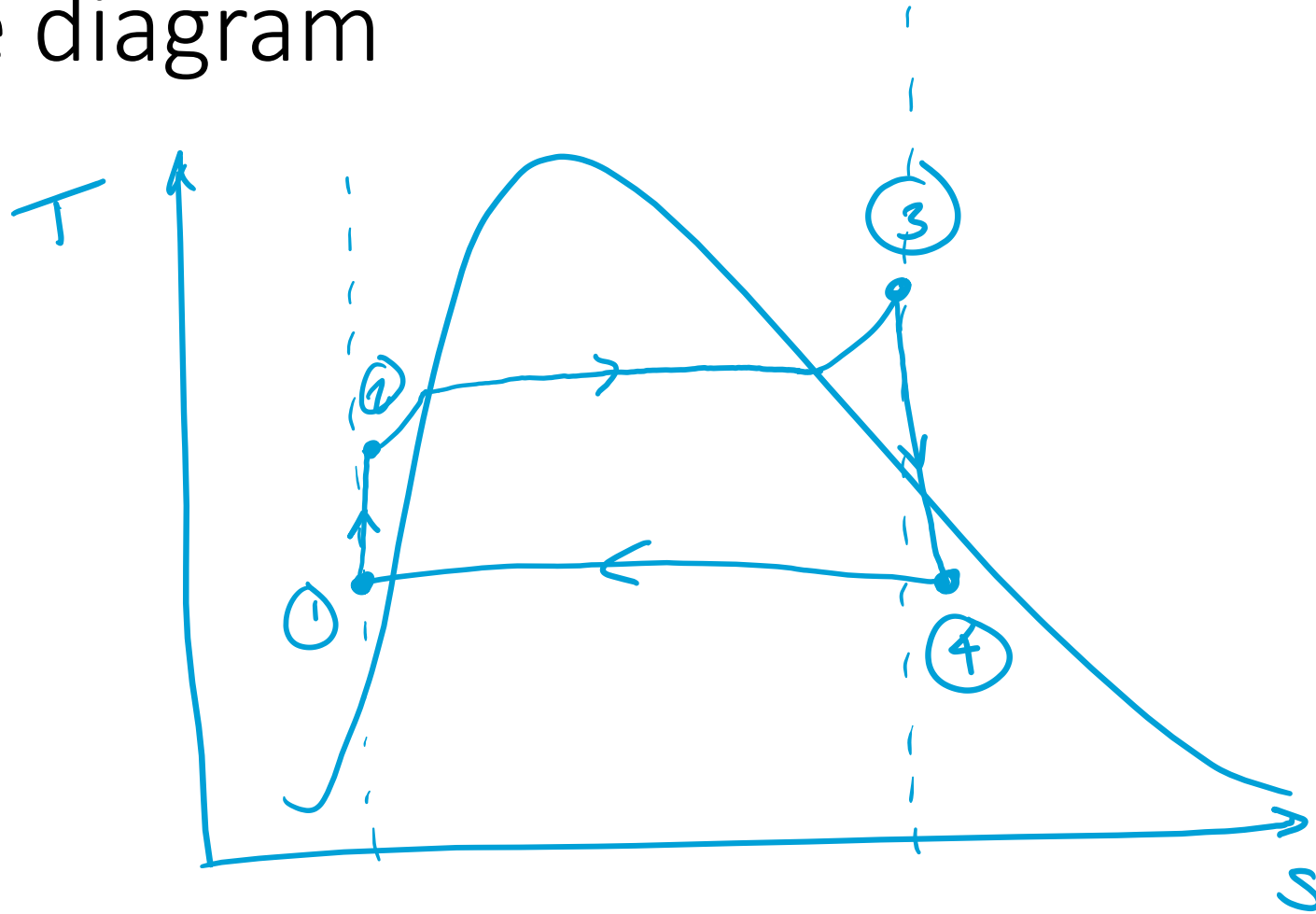
$h_{4,s} = x_{4,s} h_g + (1 - x_{4,s}) h_f = 2600.5 \frac{\text{kJ}}{\text{kg}}$

State 4 : $\eta_{\text{turb}} = \frac{\dot{W}_{\text{real}}}{\dot{W}_{\text{ideal}}} = \frac{h_4 - h_3}{h_{4,s} - h_3}$

$\Rightarrow h_4 = \eta_{\text{turb}} (h_{4,s} - h_3) + h_3$

$\boxed{h_4 = 2688.308 \frac{\text{kJ}}{\text{kg}}}$

Cycle diagram



Work and heat

$$\dot{W}_{\text{pump}} = -\dot{m}(h_2 - h_1) = -25(94.18 - 70.7) = -584.3 \text{ kW}$$

$$\dot{W}_{\text{turb}} = -\dot{m}(h_4 - h_3) = -25(2688.3 - 3854.9) = 29215 \text{ kW}$$

$$\dot{Q}_{\text{in}} = \dot{m}(h_3 - h_2) = 25(3854.9 - 94.18) = 94019 \text{ kW}$$

$$\dot{Q}_{\text{out}} = \dot{m}(h_1 - h_4) = 25(70.7 - 2688.3) = 65386 \text{ kW}$$

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{pump}} + \dot{W}_{\text{turb}}}{\dot{Q}_{\text{in}}} \Rightarrow \boxed{\eta_{\text{th}} = 0.3}$$